

Luke Liu

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SUMMARY

- A seasoned Chemical & Environmental Engineer/PE/PhD with 5-year strong hand-on industrial experience and 14-year research experience
- Expertise in oil & gas production chemistry, process engineering, water/air pollution control, environmental compliance
- Excellent written and verbal communication skills
 - 10+ published journal papers; 10+ reports prepared for EPA lab and S&P 500 energy companies
- Green Card Holder (US Permanent Resident)

PROFESSIONAL SKILLS

- Lab: 14-year lab operation & management experience; chemical analysis including GC/MS, HPLC, ICP, FTIR, XRD, SEM, AFM, Raman, DLS, UV-vis, and other instruments
- Software: AutoCAD, FORTRAN, Mathematica, MODFLOW, Visual MINTEQ and Microsoft Office
- Design: Water & wastewater treatment plant, water distribution & sewage collection, civil engineering structure

PROFESSIONAL EXPERIENCE

Shale Water Research Center, LLC

Chemical Engineer

3/2014 – 12/2014

Houston, TX

- Resolved the Long Time Challenges of Production Flow Assurance from Friction Reducer (FR) Additives (Devon Energy's Project)

Challenges:

- FRs, e.g., polyacrylamide (PAM), form aggregates that decrease the oil transportation efficiency and impair the production

My Contribution/Achievements:

- Identified that FR polymer bridged with multivalent Fe^{3+} and Ca^{2+} ions is the root cause for the loss of oil transportation efficiencies (through dissolution of field aggregates)
- Successfully reproduced FR aggregates using PAM and fracturing water in lab condition
- Developed the method of eliminating field FR aggregates using citric acid through chelation of metal ions; **field production was exponentially boosted after citric acid treatment**
- Successfully prevented aggregate formation in field by applying cationic/ nonionic instead of anionic FRs
- Prepared and issued several detailed reports which had generated vast interest from shale oil companies in Texas, e.g., Apache Corporation, Devon Energy

- Barite (BaSO_4) Scale Inhibition during Long Term Storage of Fracturing Water (Apache Corporation's Project)

Challenges:

- Barite scale negatively impacts the transportation efficiency and threatens safety due to accumulation of **Naturally Occurring Radioactive Materials (NORM)**.

My Contribution:

- Investigated efficiencies of varying scale inhibitors in novel laser test to screen the most effective one
- Discovered a scale inhibitor with **20% regular dosage** and **4 times higher efficiency** than the popularly applied DTPMP by designing and conducting 5 week tests
- **Saved millions of USD per year for inhibitor consumption** (only considering usage in North Texas area)

- Formation Damage in FR's Brine Solution under Reservoir Conditions (i.e., 6000 psi and 80°C)

My Contribution:

- Discovered pressure buildup due to blockage of shale porosity and decreased conductivity in column test
- Released Fe^{3+} and Ca^{2+} from shale could interact with anionic FRs leading to formation damage

- FR Efficiency Investigations in Flow Loop System

Challenges:

- The efficiency of FRs decreased in complex compositions of fracturing water

My Contribution:

- Developed a procedure for FR loop test to screen the most effective FRs at optimal dosage
- Led in Technician Training on Water Quality Analysis (National Oilwell Varco)

Washington University in St. Louis**Research Scientist****5/2013-2/2014****St. Louis, MO**

- Environmental Interfacial Chemistry of Nanoparticles in Natural and Engineered Environment

Challenges:

- CeO₂ nanoparticles as diesel fuel additive have entered the environment and contaminated water source.

Responsibilities:

- Designed and conducted CeO₂ flocculation test; identified the surface reaction under anoxic environment

Achievements:

- Discovered the decreased flocculation efficiency of CeO₂ due to redox reactions with Fe²⁺ ions
- Enhanced removal efficiency of nanoparticles by adjusting pH, increasing ionic strength or stirring speed
- Geologic Carbon (CO₂) Sequestration, CO₂ Storage, and the Application in Enhanced Oil Recovery

U.S. Environmental Protection Agency (EPA)**Environmental Engineer****10/2009-4/2013****Ada, OK**

- Environmental Behavior and Safety of Engineered Nanomaterials (\$1.5 M)

Challenges:

- Widely used nanomaterials enter water environment and threaten the health and safety

Responsibilities:

- As a team leader, investigated water treatment efficiency of emerging nanomaterials to assist decision makers in assessing and managing their risk

Achievements:

- Discovered the decreased flocculation efficiency of TiO₂ nanoparticles due to surface impurity compositions
 - Optimized synthesis process of nanomaterials for control of the environmental hazard
 - Revealed the interception of CeO₂ nanoparticles through Fe and Al containing sediments in filtration process
 - USEPA Scientific and Technological Achievement Awards in 2012 & 2013 for noteworthy efforts
 - Perfluorooctanoic acid (PFOA)/ Perfluorooctanesulfonic acid (PFOS) Contaminated Site Remediation
- Challenges:
- **PFOA and PFOS are very toxic and not biodegradable.**
- Achievements:
- Discovered that **the release of PFOA/PFOS is controlled by electrostatic sorption on iron oxide minerals, not by sorption to organic carbon**
 - Investigation on Procedures and Requirements of EPA Application
 - Investigated major environmental statutes (e.g., CAA, RCRA, CWA) and environmental compliance
 - Prepared of quality control/ quality assurance (QA/QC) and standard operation procedure (SOP) reports
 - Contributed to completion of an EPA's nanotechnology report on nanomaterials' risk assessments (2013).

Stevens Institute of Technology**Research Assistant****01/2006-09/2009****Hoboken, NJ**

- Flocculation and Filtration of Energetic Nanomaterials in Water Solutions (funded by US Department of Defense)

Challenges:

- Energetic nanomaterials in solid fuel spread into the environment and are toxic to organisms due to small size.

Responsibilities:

- Conducted experiment and theoretical modeling on flocculation and filtration of energetic nanomaterials

Achievements:

- Discovered the solution chemistries and hydraulic factors determined removal efficiency of the boron and silicon nanomaterials by flocculation and filtration

- **Quantitatively** predicted the flocculation and filtration of the nanomaterials in treatment process
- Initially identified the nanoparticle flocculation structure in natural organic polymer solutions by use of multiple novel characterization technologies
- Publication was ranked **Top 25 Hottest Articles** in the Journal of Colloid and Interface Science, 2008
- Soil and Groundwater Remediation on Volatile Organic Compounds (VOCs) and Cr(VI) Contaminated Site Challenges:
 - Metallurgical industry site contaminated by toxic VOCs (e.g, TCE) and Cr(IV) is dangerous to the public
- My Contribution/Achievements:
 - Designed a treatability test using newly created nonconventional zero valent iron suspensions
 - Both contaminants reduced to satisfactory levels by *in situ* remediation technology
- Environmentally Friendly Ammonia Test Procedure on the Tensile Residue Stress and Corrosion of Copper Alloys

East China University of Science & Technology Research Assistant 01/2004-11/2005 Shanghai, China

- Shanghai Petrochemical Wastewater Treatment Challenges:
 - Refractory chemicals in petrochemical industry wastewater cannot be effectively treated by regular biological treatment process
- Responsibility:
 - Designed and conducted experiment by advanced oxidation process & biodegradation
- Achievements:
 - Biodegradability was greatly improved by using Fenton's reagent due to degradation of refractory chemicals
 - The effluent of following Sequence Batch Reactor satisfied regulatory requirement
 - Large amount of formerly consumed **dilution water (1.25 million m³ annually) was saved**
- Purification of TCE and PAHs Contaminated Water by Innovative Coupled Ultrasound-UV Process

Harbin Institute of Technology Research Assistant 08/2001-12/2003 Harbin, China

- Municipal Wastewater Treatment and Water Reuse in Shengli Oilfield Challenges:
 - The effect of varying media on performance of biological aerated filter (BAF) was not clearly understood.
- Achievements:
 - Discovered the surface area and structure of BAF media impacted the microorganism amount, hydraulic transport of contaminants and O₂, and therefore, the treatment efficiency of BAF.
 - Discovered the efficiency of BAF was higher than the existing wastewater treatment plant's wetland process
 - Successfully scaled up to a full-scale BAF treatment facility
- Combined Membrane Technologies (Continuous Micro-Filtration & Reverse Osmosis) in Treatment of the Effluent of Wastewater Treatment Plant in Oilfield for Reuse

CERTIFICATION

- Registered Professional Engineer (PE #26501, in Environmental Engineering)
- OSHA Hazardous Waste Operations and Emergency Response (40 h HAZWOPER, #0279)

EDUCATION

- **PhD in Environmental Engineering**, Stevens Institute of Technology, Hoboken, New Jersey, USA 2009
(Courses and dissertation focused on **colloidal & aquatic chemistry**)
- **M.S. in Environmental Engineering**, Harbin Institute of Technology, Harbin, China 2003
- **B.E. in Civil & Environmental Engineering**, Harbin Engineering University, Harbin, China 2001

VOLUNTEER ACTIVITIES

- Have been a reviewer for US EPA's Science to Achieve Results (STAR) grants, Portuguese Science Foundation, and many prestigious journals such as Environmental Science & Technology, Water Research, and other journals
- **Associate Editor, Water Quality, Exposure and Health (Springer)**

SELECTED PUBLICATIONS

1. Zhao F.; Yu Z.; Park, H. D.; Liu, X.; Song, X.; Li, Z. Polyvinylchloride Ultrafiltration Membranes Modified with Different SiO₂ Particles and Their Antifouling Mechanism for Oil Extraction Wastewater. **Journal of Environmental Engineering**. (In Press)
2. Yu Z.; Liu, X.; Zhao F.; Liang X.; Tian Y. Fabrication of a low cost nano SiO₂/PVC composite ultrafiltration membrane and its antifouling performance. **Journal of Applied Polymer Science**. 2015, 132, 41267.
3. Liu, X.; Chen, G.; Erwin, J.; Su, C. Silicon Impurity Release and Surface Transformation of TiO₂Anatase and Rutile Nanoparticles in Water Environments. **Environmental Pollution**. 2014, 184, 570-578.
4. Liu, X.; Chen, G.; Erwin, J.; Adam, N.; Su, C. Release of Phosphorous Impurity from TiO₂Anatase and Rutile Nanoparticles in Aquatic Environments and Its Implications. **Water Research**. 2013, 47, 6149-6156.
5. Liu, X.; Chen, G.; Keller, A.A.; Su, C. Effects of dominant material properties on the stability and transport of TiO₂ nanoparticles and carbon nanotubes in aquatic environments: From synthesis to fate. **Environmental Science: Processes & Impacts**. 2013, 15 (1), 169 - 189. (Invited)
6. Liu, X.; Chen, G.; Su, C. Influence of Collector Surface Composition and Water Chemistry on the Deposition of Cerium Dioxide Nanoparticles: QCM-D and Column Experiment Approaches. **Environmental Science & Technology**. 2012, 46, 6681-6688.
7. Chen, G.; Liu, X.; Su, C. Distinct Effects of Humic Acid on Transport and Retention of TiO₂ Rutile Nanoparticles in Saturated Porous Media. **Environmental Science & Technology**. 2012, 46, 7142-7150.
8. Ferrey, M.L.; Wilson, J. T.; Adair, C.; Su, C.; Fine, D. D.; Liu, X.; Washington, J.W. Behavior and Fate of PFOA and PFOS in Sandy Aquifer Sediment. **Ground Water Monitoring & Remediation**. 2012, 32, 63-71.
9. Liu, X.; Chen, G.; Su, C. Effects of material properties on sedimentation and aggregation of titanium dioxide nanoparticles of anatase and rutile in the aqueous phase. **Journal of colloid and interface science**. 2011, 363, 84-91.
10. Liu, X.; Wazne, M.; Chou, T.; Xiao, R.; Xu, S. Influence of Ca²⁺ and Suwannee river humic acid on aggregation of silicon nanoparticles in aqueous media. **Water Research**. 2011, 45, 105-112.
11. Chen, G.; Liu, X.; Su, C. Transport and retention of TiO₂ rutile nanoparticles in saturated porous media under low-ionic-strength conditions: Measurements and mechanisms. **Langmuir**, 2011, 27, 5393-5402.
12. Liu, X.; Wazne, M.; Christodoulatos, C.; Jasinkiewicz, K. Effects of natural organic matter on aggregation kinetics of boron nanoparticles in monovalent and divalent electrolytes. **Journal of colloid and interface science**. 2010, 348, 101-107.
13. Liu, X.; Wazne, M.; Christodoulatos, C.; Jasinkiewicz, K. Aggregation and deposition behavior of boron nanoparticles in porous media. **Journal of colloid and interface science**. 2009, 330, 90-96. *
14. Strigul, N.; Vaccari, L.; Galdun, C.; Wazne, M.; Liu, X.; Christodoulatos, C.; Jasinkiewicz, K. Acute toxicity of boron, titanium dioxide, and aluminum nanoparticles to *Daphnia magna* and *Vibrio fischeri*. **Desalination**. 2009, 248, 771-782.
15. Liu, X.; Du, M.; Fan, Z.; Wang, H.; Han, H. Research on performances of media in wastewater treatment by biological aerated filter. **Journal of Harbin Institute of technology**. 2006, 36, 1835-1839. (In Chinese)

* **Top 25 Hottest Articles** in the Journal of Colloid and Interface Science, October to December, 2008